

Rates



1 Tricia reads 20 pages of her book in 2 hours. Find her reading rate of pages per hour.

2 A car travels 320 km in 4 hours.

a Complete the following table of values:

Time taken (hrs)	4	2	1
Distance travelled (km)	320		

b Find the speed of the car in kilometres per hour.

3 A tap fills up a 240 litre tub in 4 hours.

a State the compound unit for the rate of water flow.

b Find the rate of water flow of the tap.

4 A toy manufacturer makes 720 toys in 9 hours.

a State the compound unit for the rate of toy manufacturing.

b Find the rate at which the toy manufacturer works.

5 Justin earns \$420 in 6 hours. Find Justin's hourly rate.

6 Tina burns 640 calories by jogging 8 kilometres. Find the rate at which Tina burns calories per kilometre.

7 Luke writes 900 words in 20 minutes. Find Luke's writing speed in words per minute.

8 A 3-episode TV series had a total run time of 69 minutes. Find the run time for a single episode.

9 An animator uses 12 frames for every 0.5 seconds of an animation. Find the frame rate of the animation in frames per second (fps).

10 Avril is running from John in a game of tag. Avril runs 30 metres every 5 seconds and John runs 21 metres every 3 seconds.

a Find Avril's speed in metres per second. b Find John's speed in metres per second.

c Who is faster?



Applications

- 11** Iain earns \$22 per hour working as a performer and works for 20 hours a week. Find Iain's weekly earnings.
- 12** Caitlin plays cricket over the summer holidays. Caitlin scores at a rate of 24 runs per game and there are 30 games in the season. How many runs did Caitlin score in total?
- 13** A bus travels at a speed of 80 km/hr. How far does the bus travel in 5 hours?
- 14** It requires 8 mL of polyethylene to create 20 plastic bags.
- a** Find the rate of polyethylene used per plastic bag.
 - b** How much polyethylene would be required to create 5 plastic bags?
- 15** A worm takes 15 seconds to travel 20 cm.
- a** How long does it take for the worm to travel 1 metre?
 - b** Find the rate of the worm's time spent per metre distance traveled.
- 16** A tap is dripping at a rate of 1000 mL/hr . A 5 L bucket is placed underneath it to catch the water.
- a** How many millilitres of water can the bucket hold?
 - b** How many hours will it take to fill the bucket?
- 17** Lisa is trying to save money so she deposits \$3.20 into her bank account for every 8 coffees she purchases.
- How much does Lisa deposit for each cup of coffee?
- 18** Patricia eats 7.2 litres of ice cream in 6 minutes in an ice-cream eating contest. Patricia wants to find her rate of ice-cream consumption in millilitres per second.
- a** How many millilitres of ice-cream did Patricia consume?
 - b** How many seconds did it take for Patricia to consume the ice-cream?
 - c** Find her rate of consumption of ice-cream in mL/s.
- 19** A cyclist travels 70 metres per 10 seconds. Find the speed of the cyclist in metres per minute.

20 Tina counts 26 heart beats over a 20 second interval. Find Tina's heart rate in beats per minute (bpm).



21 Adam really likes apples and eats them at a rate of 4 per day.

- a** How many apples does Adam eat in one week?
- b** If Adam buys 44 apples how many days will this last him?

22 The height of a pool increases by 5 cm for every 10 litres of water added. After adding some water the height of the pool increased by 0.45 m.

How much water was added to the pool?